# Re-examination Quantum Physics (GEMF)

*Coen de Graaf — Departament de Química Física i Inorgànica, URV*

---

- The exam has 6 questions, each question is worth $\frac{5}{3}$ points.
- There are 8 pages, all sheets are printed on both sides.
- Answer each question on a separate sheet, clearly mark the sheets with your name.
- When asked for a numerical answer, always demonstrate how you got the number. Just giving the number will not be enough to get the maximum points.
- The maximum time to answer the questions is four hours.

## Good luck

1. **Stern Gerlach experiment**

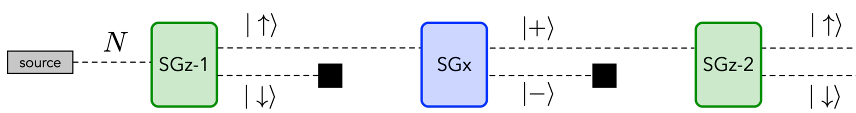

The left side of the picture above gives a schematic representation of the Stern-Gerlach experiment. The source sends an electron in the quantum state

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \quad \text{or} \quad |\psi_2\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$$

through a magnetic field along the $z$-axis (SGz) to give rise to two possible outcomes, one with the electron in the $|\uparrow\rangle$ quantum state and one with the electron in the $|\downarrow\rangle$ quantum state. The right side of the picture is similar but now the magnetic field is aligned along the $x$-axis (SGx). The electron that leaves the magnetic field is either in quantum state $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ or $|-\rangle = (|\uparrow\rangle - |\downarrow\rangle)/\sqrt{2}$.

(a) Write down the quantum states $|\psi_1\rangle$, $|\psi_2\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|+\rangle$ and $|-\rangle$ as column vectors.

In a new experimental setup, the electrons in quantum state $|\downarrow\rangle$ are blocked by the black square in the figure below and only the electrons in the $|\uparrow\rangle$ quantum state reach the second magnetic field, which is aligned along the $x$-axis. After leaving the second magnetic field, only the electrons in quantum state $|+\rangle$ can reach the last magnetic field, aligned along $z$ again.




(b) What is the probability that an electron that leaves the second magnetic field (SGx) is in quantum state $|+\rangle$ or in quantum state $|-\rangle$?

(c) What is the probability that an electron that leaves the third magnetic field (SGz-2) is in quantum state $|\uparrow\rangle$ or in quantum state $|\downarrow\rangle$?

2. **Two-dimensional harmonic oscillator**

The Hamiltonian of a two-dimensional harmonic oscillator is given by

$$\hat{H} = \frac{-\hbar^2}{2m}\left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2}\right) + \frac{1}{2}m\omega^2(x^2 + y^2)$$

(a) Find the energy eigenvalues by the *separation of variables* technique

(b) Give the degeneracy of each energy level

---

The energy eigenvalues of the one-dimensional harmonic oscillator are

$$E_n = (n + \frac{1}{2})\hbar\omega, \text{ with } n = 0, 1, 2, 3, \ldots$$

3. **Eigenfunctions, eigenvalues and expectation values**

(a) Given the normalized wave functions $\psi = Ae^{-3x^2}$ and $\phi = Bxe^{-3x^2}$, and the operator

$$\hat{Q} = \lambda \frac{d^2}{dx^2} - 36\lambda \hat{X}^2 \qquad (\lambda \in \mathbb{R})$$

associated to the observable $q$, show that $\psi$ and $\phi$ are eigenfunctions of $\hat{Q}$.

(b) What are the possible outcomes of measuring $q$ for a quantum system described by $\psi$ or by $\phi$?

(c) Determine the normalization constant $\mu$ of $\theta = \mu(\psi + \sqrt{3}\phi)$

(d) Show that $\theta$ is not an eigenfunction of $\hat{Q}$

(e) What are the possible outcomes of measuring $q$ for a quantum system described by $\theta$?

(f) What is the average outcome of measuring $q$ in $N$ systems in quantum state $\theta$ ($N \to \infty$)?

4. **Perturbation theory with matrices**

The Hamiltonian of a quantum system is defined as

$$\hat{H} = \begin{pmatrix} 4 & 1 & 1 \\ 1 & -2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

Since the exact determination of the eigenvalues is rather involved they will be approximated with perturbation theory.

(a) Write down the perturbation operator $\hat{V}$ in matrix format when the zeroth-order Hamiltonian is given by

$$\hat{H}^{(0)} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(b) Give the zeroth-order eigenvalues $E_1^{(0)}$, $E_2^{(0)}$, $E_3^{(0)}$ and the corresponding zeroth-order eigenvectors.

(c) Calculate the first-order correction for the three energy eigenvalues.

(d) Calculate the second-order correction for the three energy eigenvalues.

(e) Calculate the energy eigenvalues adding the first- and second-order corrections to the zeroth-order energy eigenvalues and compare to the exact eigenvalues.

---

The three exact eigenvalues are

$$E_1 \approx 4.546 \qquad E_2 \approx -2.169 \qquad E_3 \approx 1.623$$

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{V} | \psi_n^{(0)} \rangle \qquad E_n^{(2)} = \sum_{m \neq n} \frac{|\langle \psi_n^{(0)} | \hat{V} | \psi_m^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

5. **Hydrogen-like atoms**

The figure on the next sheet shows the radial part of some of the eigenfunction of the hydrogen atom (left) and the corresponding probability distribution next to it.

(a) Calculate the most probable radius for the electron when the hydrogen atom is in its ground state.

$He^+$, $Li^{2+}$ and $Be^{3+}$ are examples of hydrogen-like atoms: one electron interacting with a nucleus of charge $+Z$.

(b) Calculate the most probable radius for the electron in the ground state of $He^+$ ($Z = 2$) and $Li^{2+}$ ($Z = 3$) and extrapolate the result for a general hydrogen-like atom with nuclear charge $Z$.

(c) Show that the most probable radius for an electron in a hydrogen-like atom described by the wave function $R_{n,l=n-1}$ is given by $n^2/Z$

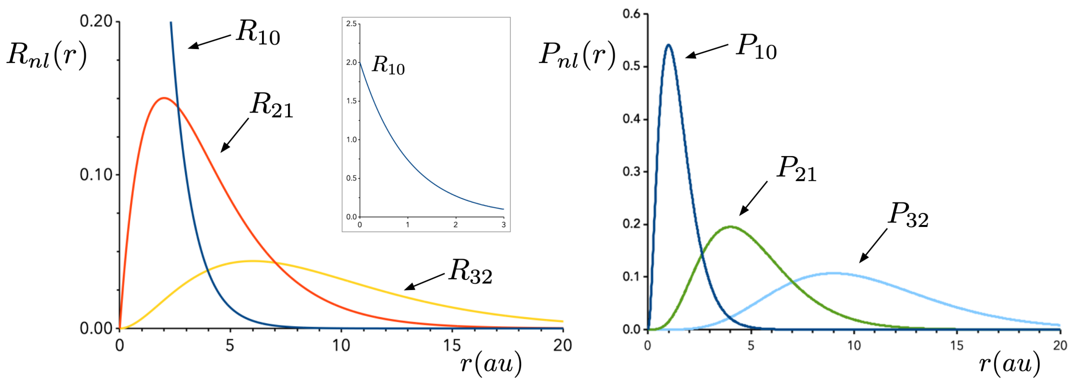
---

Hydrogen atom: $R_{10} = 2e^{-r}$ Hydrogen-like atoms: $R_{10} = 2Z^{3/2}e^{-Zr}$

Other useful functions for hydrogen and hydrogen-like atoms:

$R_{21} = Are^{-Zr/2}$ $R_{32} = Ar^2e^{-Zr/3}$

Remember: $dV = r^2 dr d\Omega$



$R_{nl}(r)$
$R_{10}$
$R_{21}$
$R_{32}$
$r(au)$
$P_{nl}(r)$
$P_{10}$
$P_{21}$
$P_{32}$
$r(au)$

6. **Matrix mechanics**

The $\hat{N}$ operator of a four-dimensional Hilbert space

$$\hat{N} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \tag{1}$$

has the following eigenvalues and eigenstates:

$$n_i = 0, 1, 2, 3 \quad \text{and} \quad |n_i\rangle = |0\rangle, |1\rangle, |2\rangle, |3\rangle$$

Note that $|0\rangle$ does not represent a zero quantum state (vacuum), but rather the quantum state with the lowest eigenvalue of $\hat{N}$.

(a) Construct the matrix representation of the annihilation and creation operators $\hat{a}$ and $\hat{a}^\dagger$. Each matrix element $(i, j)$ corresponds to $\langle n_i|\hat{a}|n_j\rangle$ or $\langle n_i|\hat{a}^\dagger|n_j\rangle$, where $i$ is the $i^{th}$ row and $j$ the $j^{th}$ column.

(b) Does $\hat{a}$ commute with $\hat{a}^\dagger$?

(c) Show that $\hat{N} = \hat{a}^\dagger \hat{a}$

(d) Write the eigenvectors of the number operator $\hat{N}$ as column vectors and show that

$$\langle n_i| \hat{a}^\dagger = \langle n_{i-1}| \sqrt{n} \quad \text{and} \quad \hat{a} |n_i\rangle = \sqrt{n} |n_{i-1}\rangle$$

and hence

$$\langle n_i| \hat{a}^\dagger \hat{a} |n_i\rangle = n$$

(e) Are the eigenvectors of $\hat{N}$ also eigenvectors of $\hat{a}$ and $\hat{a}^\dagger$?

---

$$\hat{a} |n\rangle = \sqrt{n} |n_{i-1}\rangle \qquad \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n_{i+1}\rangle$$